Status	Finished
Started	Thursday, 29 October 2026, 5:52 PM
Completed	Thursday, 29 October 2026, 5:55 PM
Duration	3 mins 1 sec
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It only works on small datasets
- c. It avoids the need for recursion
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **2**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. order()
- c. sorted()
- d. arrange()

The correct answer is: sort()

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Merge Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Bubble Sort
b. Insertion Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It has no effect on the search operation
c. It allows for faster searching
d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **6**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are combined to form a single sorted list
b. They are compared element by element
c. They are split again into smaller sublists
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **7**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data in a single pass
b. Dividing the input into parts, solving each part, and combining the solutions
c. Rearranging data without sorting
d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **8**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Bubble Sort
c. Linear Search
d. Merge Sort

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items
b. It slows down the process
c. It makes it easier to select items based on their relationship to the rest of the items
d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **10**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It makes the analysis slower
c. It allows for quicker identification of duplicates
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Quick sort
c. Merge sort
d. Selection sort

The correct answer is:
Merge sort

Question **12**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. All of the above
b. Selection
c. Duplicates
d. Searching

The correct answer is: All of the above

Question **13**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Insertion sort

b. Heap sort

c. Bubble sort

d. Quick sort

The correct answer is:
Insertion sort

Question **14**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470

b. 40,80,90,150,200,300,400,470,70,120

c. 200,470,80,150,40,90,300,400,70,120

d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **15**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sorted()

b. sort()

c. arrange()

d. order()

The correct answer is: sorted()